

Mixed Reality Scenarios for Speech Therapy: Report

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ABSTRACT

This project aims to create a virtual reality (VR) tool to aid in speech therapy. This tool was motivated by the fact that these patients are unable to apply the techniques learned in traditional speech therapy when faced with stressful real-world situations. Our project was the bridge that was needed to link the therapy office to reality. Our project was based on a school auditorium in which it asked the patient to recite a speech in front of a simulated classroom. To further induce realism, we added real recordings of sounds that were collected in an auditorium. Further testing is needed with real patients to properly gauge stress levels and the success of this tool in therapy; current observations are among users without speech disorders.

KEYWORDS

Virtual Reality, Speech Therapy, Real-world Simulation

1 INTRODUCTION

Within the United States, millions of children and adults have reported that they have stutters, aphasia, and/or other speech impairments [1]. There are many methods that speech therapists employ for their patients depending on what speech disorder they are suffering from. This could include playing with a child who is a late talker to encourage them to speak, exercises to control the rate of speech for people that suffer from stuttering, and speech drills to aid individuals with aphasia [5]. Even though these techniques work well for some individuals, therapists and researchers have noticed that even when these patients do well with these exercises during therapy, they see less success in stressful situations in the real world and may be unable to employ the same techniques.

This is where our project closed the gap between reality and therapy. Current development in this field is leaning toward the realm of virtual reality. This is the case because therapists often teach and perform these techniques in the

office which do not allow “opportunities for real-life interaction [1]. This is a key factor that will push speech therapy to greater success. This is the case because using these real-life scenarios within the therapy session will allow the patient to better associate these techniques with interactions that they have experienced before. Another benefit of virtual reality as Ms. Danielle Aldridge, a speech pathologist working with the team at RECOVER, states is there is the possibility of a redo [1]. Patients can keep returning to a scenario that they are having trouble in until they see success in their speech which cannot be done as simply in real life.

In our project, we seek similar goals and motivations as other researchers exploring the benefits of virtual reality. We created a virtual reality environment that attempts to induce the same intense feelings of giving a speech in front of a classroom/audience. We chose this environment because it was the most practical to employ. Further development can follow with more environments to allow for variety within these therapy sessions, such as the proposed café and mock interview environments. With this environment, the patient can give a mock speech in front of an audience while employing techniques that are taught by the speech therapist in hopes of being able to employ these techniques when put into an analogous situation in real life.

2 BACKGROUND

VR has already shown promise as a therapeutic tool. Existing research has already shown positive outcomes when using VR for exposure therapy, simulating situations for symptom assessment, and training. This last use is most important when discussing speech therapy [6, 7]. Our project is not the first to explore the use of virtual environments in speech therapy. One of these programs was EVA Park, a virtual world specifically designed for people with aphasia to “receive therapy and practice communication skills” as well as express themselves in a unique way that might be more accessible than in the real world [3]. There have been studies that have shown that using a VR system has helped with treatment for aphasia or increased the salience of situations for people who stutter. A study showed that having patients

with Broca's aphasia work with a rehabilitation gaming system resulted in improvements in communication skills [4]. Another study had patients undergoing speech therapy for a stutter use a VR system, and they reported similar emotions to real-life situations, showing that the therapy was immersive and engaging [2]. What sets our project apart is that it is specifically designed to be a treatment tool for stuttering, used for both assessment and practice alongside a speech therapist.

3 IMPLEMENTATION

At the beginning of the semester, we had plans to create three different scenes that imitated real-life situations to be used for speech therapy sessions. The three scenes were someone giving a speech in a lecture hall/auditorium, ordering food at a café, and participating in a job interview, which are shown in figure 1. Unfortunately, due to course time constraints, our inability to meet with the UGA researchers spearheading this project until halfway through the semester, and our initial lack of experience with Unity and 3D modeling, we had to reduce the number of scenes created and focus on one less interactive scene.

After receiving feedback for our first iteration, a café environment, we chose to pursue an auditorium environment for our final prototype. This pivot allowed us to create a more realistic scene due to the background audio collected and the lack of one-on-one interactions with non-playable characters (NPCs).

In order to implement this VR scene, we used Unity as a 3D modeling program. Our main focus was to create a scene that was as realistic as possible given our time constraints that included features aimed at increasing the stress of the user. The first step in this process was to create the auditorium environment using premade assets in Unity. We purchased an asset package that included the auditorium environment and then added lighting, additional furniture, and other background items. Next, we wrote and implemented code to allow the user to move around the environment and experience the auditorium as a VR scene.

Once the overall auditorium environment was complete, we were able to add our NPC audience members. The first iteration café scenario featured an NPC that we coded ourselves that's result was too cartoonish for our more realistic goals, so we needed to change directions here as well. One possible alternative was to take videos of ourselves and other individuals and import them directly into our environment as flat videos. We tested this method by taking a few videos of ourselves and importing them over the seats looking toward the auditorium stage, but the results were not ideal. Even when the user only moved around on the stage where a lecturer would stand to give their speech, it was clear that

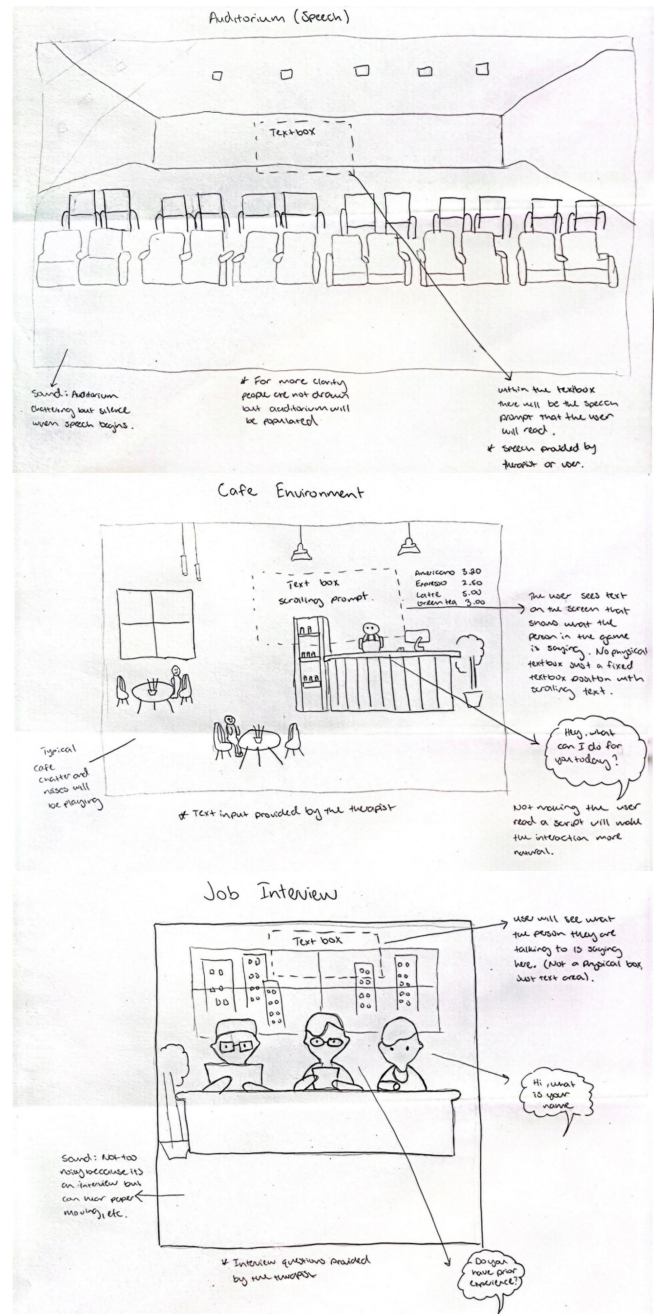


Figure 1: Design of the Three Scenes

the audience members were just 2D renderings of real people as you could see the empty space behind them. This made it obvious that the audience members were not 3D like the rest of the environment. The option we settled on involved finding a Unity asset package that contained more realistic NPCs. We found a package that contained people dressed in business or casual attire that fit our purposes and using Unity

transform and animation functions were able to change their positions, so they were seated in the auditorium like a real audience.

The last components that we implemented were background audio, scrolling text, a timer, and audience responses after the speech to make the scene more realistic. Figure 2 shows our auditorium implementation. We collected the background audio by recording ourselves in an empty classroom while making distracting and stress-inducing noises. Some of these sounds included doors opening and shutting, individuals coughing, laughing, tapping, and whispering, as well as snippets of video playing on a laptop.

The flow for our VR speech therapy scenario is designed to mimic a real-life public speaking experience in an auditorium. Initially, the user is spawned in a corner of the auditorium, representing the anticipation and the preparatory phase before a speech. The user is then prompted to navigate towards the speaker table. Upon reaching the area, the main event of the scenario commences a one-minute speech. The speech content is pre-generated by artificial intelligence, ensuring a diverse range of speeches for different sessions. This part of the scenario aims to recreate the actual experience of public speaking, complete with the pressure of delivering an articulate and coherent speech within a limited timeframe. After the completion of the speech, to simulate a real audience's reaction and provide a sense of closure and achievement, the VR environment erupts into cheers and applause. This is designed to add a layer of realism and to reinforce positive feedback to the user after their speech.

To implement the speech portion of the scenario, we coded a scrolling text box that appears once the user walks up to a podium at the front of the stage. To create actual text, we generated random speeches using ChatGPT and then added each of those speeches along with a random number generator into our code so that the user receives a different speech each time they complete the scenario. This allows them to practice many times without getting too comfortable with one specific speech. To add an extra layer of stress, we also coded a timer that counts down in a corner of the screen while the person is speaking and stops them automatically once the time is up. Lastly, we implemented randomly generated audience responses, such as cheering or clapping, that occur after the timer has stopped.

4 RESULTS

Due to time constraints, we were not able to test our prototype on individuals with speech disorders; however, we supplemented this by doing a study on how realistic this environment was and if it could induce stress on our participants. Our participants were selected through convenience sampling to get a representative sample of individuals that



Figure 2: Auditorium VR Environment

were comfortable with public speaking and uncomfortable with public speaking. We were able to test our prototype with five participants, noting that this is not enough participants for a generalizable conclusion, but our results provide valuable insight.

We utilized two Likert scales to assess how comfortable our participants were with public speaking (Figure 3) and how realistic they felt our environment was at inducing stress (Figure 4). Comfortability was rated on a five-point scale, 1: Very Uncomfortable and 5: Very Comfortable and Realism was rated on a five-point scale, 1: Very Poor 5: Very Well. From our observations, we noticed that there was a correlation between comfortability and realism. The individuals that noted the environment was not real enough to induce stress were comfortable with public speaking, however, the environment was rated more real among the individuals that do not fond of public speaking. Also, the survey also included a section that allowed participants to provide recommendations for what can be added to include more realism. Many of the answers addressed an issue that we attempted to resolve in our project but fell short. The participants noted that the environment could have induced more stress if the NPCs looking at them were more realistic. In other words, they wanted real faces to stare back at them instead of animated characters. This was discussed during implementation, but with the time allotted could not be done. This point is noted for further implementations in which we can use still clips of real students/people.

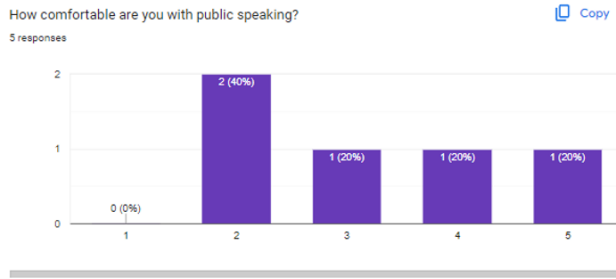


Figure 3: Result of Q1

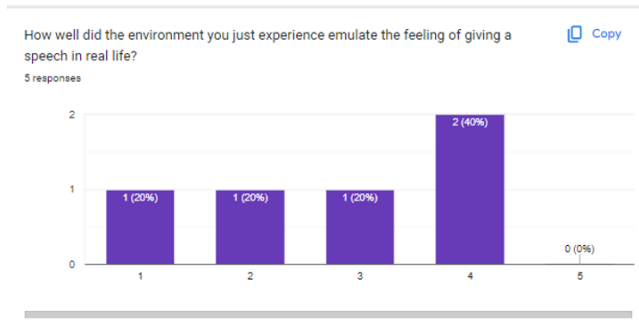


Figure 4: Result of Q2

We also intended to include feedback from the UGA researchers regarding our final prototype but were unable to do so in a timely manner.

5 CONCLUSION AND REFLECTION

For this project, we aimed to employ at least one environment that is functional and can be used for testing. We planned to make a café environment first; however, due to the complexity of that scenario in terms of assets and sound, it was difficult to make it realistic. In our first iteration (café environment) the NPC's, were too simplistic and cartoonish which would hurt the realism of the environment which directly correlates to the stress that is felt. Also, we were not able to collect enough audio from cafés to make a realistic scene. In the end, we chose to pursue the auditorium scene due to the amount of audio we could collect. Another factor in this decision was that the NPCs are a greater distance from the user so their cartoonish nature would not distract from the realism of the environment as much as a one-on-one interaction, such as in the café iteration.

Overall, we've completed the objective of creating a virtual reality environment that our UGA colleagues can use for further testing. The researchers that we are working with have access to patients with speech disorders, thus it is more appropriate for them to employ this prototype for testing.

The current implementation only has one scene out of the discussed three scenes. We chose to focus on one environment to create a prototype that is fully functional and is ready for further testing. Also, we concluded that after the base code is done for one environment the other environments should be similar in structure just varying in interaction. Therefore, further work can be done by finding scenarios that patients find uncomfortable or induce stress and implementing them in virtual reality.

REFERENCES

- [1] 2023. Virtual reality helps speech pathologists. <https://oztrekk.com/news/virtual-reality-helps-speech-pathologists/> [Online; accessed 28-July-2023].
- [2] Abeer Al-Nafjan, Najwa Alghamdi, and Abdulaziz Almudhi. 2021. Virtual Reality Technology and Speech Analysis for People Who Stutter. *EMITTER International Journal of Engineering Technology* 9, 2 (2021), 326–338.
- [3] Lucy Bryant, Melissa Brunner, and Bronwyn Hemsley. 2020. A review of virtual reality technologies in the field of communication disability: Implications for practice and research. *Disability and Rehabilitation: Assistive Technology* 15, 4 (2020), 365–372.
- [4] Klaudia Grechuta, Belen Rubio, Armin Duff, E Duarte Oller, F Pulvermuller, and PFMJ Verschure. 2016. Intensive language-action therapy in virtual reality for a rehabilitation gaming system. *J Pain Manag* 9, 3 (2016), 243.
- [5] Kristin R. Hayes. 2022. The Different Types of Speech Therapy. <https://www.verywellhealth.com/types-of-speech-therapy-1192153> [Online; accessed 28-July-2023].
- [6] K Jashwanth, Swaroop Raj Shetty, AN Yashwanth, and Roopa Ravish. 2020. Phobia Therapy Using Virtual Reality. In *2020 IEEE International Conference for Innovation in Technology (INOCON)*. IEEE, 1–6.
- [7] M Krijn, P.M.G Emmelkamp, R.P Olafsson, and R Biemond. 2004. Virtual reality exposure therapy of anxiety disorders: A review. *Clinical Psychology Review* 24, 3 (2004), 259–281. <https://doi.org/10.1016/j.cpr.2004.04.001>